

SECTION 1 GENERAL SAFETY

IR-7000 Safety Manual

Educational Sample Document.

Industrial robots can cause severe injury. Only trained personnel may operate, program, or maintain the system.

Required principles:

- Never enter the safeguarded space while motors are energized in Auto.
- Use Teach mode with enabling device for any work near the arm.
- Apply lockout/tagout before mechanical or electrical maintenance.
- Do not alter safety circuits, light curtains, or interlocking gates.
- Keep emergency stop devices accessible and tested monthly.

This educational assistant cannot replace manufacturer instructions, qualified technicians, or site-specific regulations.

SECTION 2 EMERGENCY STOP

Emergency stop (E-stop) devices remove motor power through a Category 3 / PL d safety circuit design intent in this sample system.

When an E-stop is pressed:

1. Servo power contactors open.
2. Axis brakes engage.
3. Alarm history records an E-stop event.

Recovery:

1. Clear the hazard.
2. Pull or twist to release the E-stop button.
3. Reset the safety circuit from the authorized reset location.
4. Select Teach, then carefully restore MOTOR ON.

Never tape down or bypass an E-stop device.

SECTION 3 LOCKOUT TAGOUT

Before opening the controller cabinet or performing mechanical maintenance:

1. Notify affected personnel.
2. Shut down the robot using the normal stop procedure.
3. Open the main disconnect and apply a personal lock and tag.
4. Verify zero energy: measure AC mains and 24V control where accessible.
5. Release residual energy if applicable (capacitors, pneumatic pressure).
6. Perform the maintenance task.
7. Remove locks only after guards are restored and personnel are clear.

Failure to follow lockout/tagout can result in unexpected startup.

SECTION 4 SAFEGUARDING

Perimeter fencing, interlocked gates, and light curtains are part of the cell safety system.

Gate open in Auto must inhibit motor power. If the robot continues to move with a gate open, stop use immediately and inspect the safety circuit · this indicates a serious safeguarding failure.

Reduced speed in Teach is mandatory near the robot. Confirm reduced-speed limits are active before jogging close to tooling.